DATABASE SCHEMA & DATA DICTIONARY

Core User & Content Catalog Tables

LEARNER — Registered student accounts

Column Name	Data Type	Constraint	Sample Entry 1	Sample Entry 2
PK learner_id	INT	NOT NULL	1001	1002
UQ email	VARCHAR(80)	NOT NULL	learner1@elemental.kr	minjung@topik.kr
username	VARCHAR(30)	NOT NULL	k_learner	minjung_k
password_hash	VARCHAR(255)	NOT NULL	$2b$12$e8x...	$2b$12$q9m...
target_topik_level	TINYINT	DEFAULT 1	2	4

TOPIC — Grammar and vocabulary learning modules

Column Name	Data Type	Constraint	Sample Entry 1	Sample Entry 2
PK topic_id	INT	NOT NULL	201	202
FK level_id	INT	NOT NULL	1 (TOPIK I)	2 (TOPIK II)
topic_name	VARCHAR(100)	NOT NULL	Honorific Verbs (-시-)	Cause & Effect (-아/어서)
sequence_order	INT	NOT NULL	1	2

LESSON — Structured learning materials

Column Name	Data Type	Constraint	Sample Entry 1	Sample Entry 2
PK lesson_id	INT	NOT NULL	3001	3002
FK topic_id	INT	NOT NULL	201	201
title	VARCHAR(120)	NOT NULL	Introduction to Honorifics	Formal Conversation Practice
content_body	TEXT	NOT NULL	"Use -시- when describing..."	"Dialogue practice between..."

Explanation: Tables enforce relational dependencies. Primary Keys (PK) uniquely identify rows, Foreign Keys (FK) establish structural links, and Unique constraints (UQ) prevent duplicate credentials across user profiles and learning catalogs.

DATABASE SCHEMA & DATA DICTIONARY

Assessment & Adaptive Learning Tables

QUESTION — Tier-tagged question items

Column Name	Data Type	Constraint	Sample Entry 1	Sample Entry 2
PK question_id	INT	NOT NULL	5001	5002
FK quiz_id	INT	NOT NULL	401	401
FK tier_id	INT	NOT NULL	1 (Beginner)	3 (Advanced)
question_text	TEXT	NOT NULL	"Select the correct honorific..."	"Complete the formal passage..."

QUIZ_ATTEMPT — Recorded student quiz sessions

Column Name	Data Type	Constraint	Sample Entry 1	Sample Entry 2
PK attempt_id	INT	NOT NULL	8001	8002
FK learner_id	INT	NOT NULL	1001	1001
FK quiz_id	INT	NOT NULL	401	401
score_percentage	DECIMAL(5,2)	NOT NULL	85.00	45.00
submitted_at	DATETIME	NOT NULL	2026-11-05 14:20	2026-11-08 09:15

LEARNER_PROGRESS — Adaptive state and rolling accuracy tracker

Column Name	Data Type	Constraint	Sample Entry 1	Sample Entry 2
PK progress_id	INT	NOT NULL	9001	9002
FK learner_id	INT	NOT NULL	1001	1002
FK topic_id	INT	NOT NULL	201	202
FK current_tier_id	INT	NOT NULL	2 (Intermediate)	1 (Beginner)
rolling_accuracy	DECIMAL(5,2)	NOT NULL	82.50	48.00

Explanation: The assessment schema links quiz results directly to learner progress. The system queries `LEARNER_PROGRESS` to obtain `current_tier_id`, filtering suitable `QUESTION` rows for upcoming practice sessions.